

Kevin Sanchez Ramos

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EDUCATION

University of Central Florida

Degree: Bachelor of Science in Aerospace Engineering

Graduation: Summer 2028

TECHNICAL SKILLS

CAD & Modeling: SolidWorks, Onshape, Fusion 360, Inventor, OpenRocket, Revit

Software & Tools: C Programming, Python, Microsoft Excel, Aerodynamic Flight Simulations, FreeFlyer, FDM Additive Manufacturing, Soldering

ENGINEERING EXPERIENCE & TEAM PROJECTS

AIAA Design Teams | *Aerospace Component & Production Specialist*

August 2025 – June 2026

- Directed a 6-person engineering team to design and fabricate a high-speed RC aircraft rated for 100+ mph, optimizing custom infill geometries using PLA, ABS, and Carbon Fiber composites to balance shear strength and airframe weight.
- Conducted computational airfoil simulations and wind tunnel data correlation to ensure flight-envelope stability, managing technical compliance files and purchase specifications.
- Supervised a 4-person quadcopter group in SolidWorks to model chassis arms and structural components, integrating onboard avionics and custom soldered harnesses.
- Designed and 3D printed an aerodynamic PLA nose cone verified via SolidWorks stress and flow simulations, successfully supporting launch performance beyond 1,000 feet AGL.
- Coordinated a 4-member sub-team to architect a motor-directed ejection recovery mechanism and configure integrated ground support electronics.

Knights Experimental Rocketry (KXR) | *NAR Level 1 High-Power Rocketry Designer*

December 2025 – July 2026

- Constructed and launched an L1 high-power rocket utilizing an AeroTech H182R motor, validating OpenRocket trajectories to reach 1,700+ ft altitude, a max velocity of 529 ft/s, and peak acceleration of 505 ft/s².
- Controlled project procurement within a \$150 budget, integrating a 30-inch nylon parachute, tubular Kevlar shock cords, rail buttons, and motor mount retention assemblies.
- Tested and evaluated various materials such as Polyethylene Terephthalate Glycol (PETG), Carbon Fiber (CF), Polylactic Acid (PLA), and Acrylonitrile Butadiene Styrene (ABS)

Knights Satellite Club (KSC) | *ADCS Subsystem Member*

May 2026 – July 2026

- Evaluated mechanical mounting fixtures for optical camera gimbals and evaluated attitude determination and control systems, analyzing reaction wheels and control moment gyroscopes (CMGs).

Academy of Engineering Gateway | *Civil and Architecture designer*

August 2021 – May 2024

- Developed comprehensive 2D floor plans and 3D elevations detailing structural layouts for multiple apartment units. Designed complex blueprints ensuring realistic traffic flow, structural integrity, and efficient space utilization. Incorporated civil engineering standards for essential site amenities including parking layout, utility lines, and drainage access.

INDEPENDENT PROJECTS

Autonomous GPR Mapping UAV | *Payload Drone System Designer*

June 2025 – Present

- Designing an autonomous UAV platform engineered for GPS-denied environments, integrating sensor mounts to support specialized Ground Penetrating Radar (GPR) instrumentation.

RepRapable Recyclebot | *Filament Extrusion Prototype*

December 2025 – Present

- Building a mechanical extrusion system to shred, melt, and pull recycled consumer plastics into uniform 1.75mm additive manufacturing filament to lower production costs.